LSTM Numerical Problems & Solutions

A Practical Guide to Long Short-Term Memory Architecture

Core Architecture Equations

Forget Gate: $f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$
Input Gate: $i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$
Candidate Cell State: $C_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$
Cell State Update: $C_t = f_t * C_{t-1} + i_t * C_t$
Output Gate: $o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$
Hidden State Update: $h_t = o_t * \tanh(C_t)$

Section 1: Basic Level

Question 1: Forget Gate Calculation

Given a previous hidden state h_{t-1} = 0.5 and a current input x_t = 1.0. The forget gate weights are W_f = [0.2, 0.8] and bias b_f = -0.5. Calculate the forget gate activation f_t.

Step-by-Step Solution:
1. Calculate linear combination: z = (0.2 * 0.5) + (0.8 * 1.0) - 0.5 = 0.1 + 0.8 - 0.5 = 0.4

2. Apply Sigmoid: σ(0.4) = 1 / (1 + $e^{-0.4}$) ≈ 1 / (1 + 0.6703) ≈ 0.5987

Answer: f_t ≈ 0.60

Question 2: Cell State Update

Suppose the previous cell state C_{t-1} = 2.0. Given: Forget gate f_t = 0.7, Input gate i_t = 0.5, and Candidate state $\tilde{C}_t$ = 0.8. Calculate the new cell state C_t.

Step-by-Step Solution:
1. Memory Retained: 0.7 * 2.0 = 1.4

2. New Information Added: 0.5 * 0.8 = 0.4

3. Summation: C_t = 1.4 + 0.4 = 1.8

Answer: C_t = 1.8

Section 2: Advanced Level

Question 3: Full Hidden State Computation

Calculate the final hidden state h_t given: New Cell State C_t = 0.6, Output Gate Weights W_o = [0.5, 0.5], Bias b_o = 0, and Inputs h_{t-1} = 0.2, x_t = 0.8.

Step-by-Step Solution:
1. Output Gate (o_t): σ((0.5*0.2) + (0.5*0.8) + 0) = σ(0.5) ≈ 0.6225

2. Calculate tanh(C_t): tanh(0.6) ≈ 0.5370

3. Final State: h_t = 0.6225 * 0.5370 ≈ 0.3343

Answer: h_t ≈ 0.334

Question 4: Gradient Flow (BPTT)

In a simplified LSTM ($C_t = f_t * C_{t-1}$), find $\partial L/\partial C_{t-1}$ given $\partial L/\partial C_t = 0.4$ and $f_t = 0.9$.

Step-by-Step Solution:

1. Apply Chain Rule: $\partial L/\partial C_{t-1} = (\partial L/\partial C_t) * (\partial C_t/\partial C_{t-1})$

2. Derivative: $\partial C_t/\partial C_{t-1} = f_t$

3. Calculate: $0.4 * 0.9 = 0.36$

Answer: Gradient is 0.36